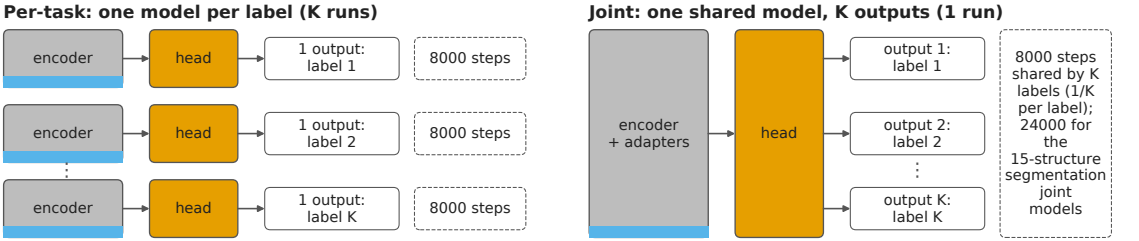

Classification loss (masked, positive-weighted BCE)
L = (1/B) Σ_b Σ_t m_bt · BCE(z_bt, y_bt; w_t): summed over the known labels of each case (m_bt = 1 if label t is known), mean over the batch (B = 1). w_t = clip(neg_t / pos_t, 0.05, 20) on the training manifest. Per-task: T = 1; joint: T = K.

Segmentation loss (Dice + BCE)
Per case b and structure c: Dice_bc (soft, smooth 1) + BCE_bc (mean over the crop's voxels), sigmoid per structure; L = mean over all (b, c) of the batch. Per-task: C = 1; joint: C = K structures, each a separate sigmoid channel.

Model selection (both modes)
Validate every 1000 steps (8 times per 8000-step run) on the selection split; keep the checkpoint with the best validation macro AUROC (classification) or mean Dice over structures (segmentation), strict improvement only; report the held-out metric at that checkpoint.